

Dual-Path Enhancement Framework for Masses and Calcifications in Mammograms: A Quantitative Evaluation of Preprocessing Techniques

Athukorala S.A.A.J.M.

*Dept. Electrical & Electronic Engineering
University of Sri Jayewardenepura
Nugegoda, Sri Lanka
janindumanjuka@gmail.com*

Deepashika S.M.D.

*Dept. Electrical & Electronic Engineering
University of Sri Jayewardenepura
Nugegoda, Sri Lanka
didulanganid@gmail.com*

Wickramasingha T.D.

*Dept. Electrical & Electronic Engineering
University of Sri Jayewardenepura
Nugegoda, Sri Lanka
tirushdumil99@gmail.com*

Fernando I.A.S.I.

*Dept. Electrical & Electronic Engineering
University of Sri Jayewardenepura
Nugegoda, Sri Lanka
senalshamika@gmail.com*

Wijewardhana U. L.

*Dept. Electrical & Electronic Engineering
University of Sri Jayewardenepura
Nugegoda, Sri Lanka
uditha@sjp.ac.lk*

Balagalla U. B.

*Dept. Electrical & Electronic Engineering
University of Sri Jayewardenepura
Nugegoda, Sri Lanka
umayabalagalla@sjp.ac.lk*

Abstract—Breast cancer is a leading cause of mortality among women and early detection using mammograms is important to increase survival rates. However, due to low contrast and noise in mammograms, image enhancement techniques are required to assist in accurate diagnosis of breast cancer. This study proposes a dual-path image enhancement framework evaluated using the publicly available INBreast dataset and compared against existing mammogram enhancement methods. The proposed mass enhancement pipeline is compared against Contrast Limited Adaptive Histogram Equalization (CLAHE), Haze Reduced Adaptive Technique (HRAT) and Magma colour mapping. The proposed calcification enhancement pipeline is compared against Gaussian and Laplacian filtering. The proposed approach achieved a 72% improvement in Contrast-to-Noise ratio (CNR) and nearly a 100% increase in Structural Similarity Index (SSIM) over CLAHE for enhancement of breast masses. According to the results for enhancement of calcifications, the proposed method obtained an improved peak signal-to-noise ratio (PSNR) by 46% highlighting its potential in assisting the process of accurate diagnosis of breast cancer using mammograms.

Index Terms—mammogram, enhancement, preprocessing, breast masses, breast calcifications

I. INTRODUCTION

Breast cancer is one of the main causes of mortality among women worldwide. In 2022, there were 2.3 million women diagnosed with breast cancer and 670,000 deaths globally [1]. The five-year survival rate for localised breast cancer is about 99% [2].

Breast cancer can be diagnosed through different imaging techniques such as mammograms, ultrasound and magnetic resonance imaging (MRI). A biopsy is performed to determine whether the abnormality is cancerous when imaging tests or a physical examination detect a suspicious change in the breast [3]. During a routine mammography-based screening, two images, namely the Cranio Caudal (CC) and Mediolateral Oblique (MLO) images are obtained. The images obtained are then examined

by radiologists to categorise them according to the Breast Imaging-Reporting and Data System (BI-RADS) [4].

Computer-Aided Diagnosis (CAD) systems assist radiologists in detecting breast cancer and classifying it into different stages [5]. However, mammograms can be challenging to interpret even using CAD systems due to factors such as low contrast, background noise, and non-coherency among regions [6]. A variety of techniques are used to improve the quality of mammograms to overcome the challenges mentioned above. Such image enhancement techniques are mainly of three types as spatial domain enhancements, frequency domain enhancements and a combination of both domains [7].

This study proposes a novel enhancement pipeline to enhance the visual quality and diagnostic interpretability of mammographic images. The proposed method uses a combination of spatial domain enhancements and colour-based fusion strategies in order to enhance both breast masses and calcifications.

II. LITERATURE REVIEW

Mammogram image enhancement plays a major role in improving the performance of machine learning and DL based breast cancer detection and classification methods. This section discusses the mammogram image enhancement methods used in the related work.

Two main types of abnormalities can be observed in mammograms, as breast masses and breast calcifications [8]. Enhancing mammograms often includes the improved visualisation of masses and calcifications.

Histogram equalisation and Contrast Limited Adaptive Histogram Equalisation (CLAHE) [9] are used extensively in enhancing mammograms. Zeng et al. [10] proposed a grey-level information histogram method for improving histogram-based image contrast enhancement. A multi-step mammogram enhancement algorithm was proposed by Avci et al. [11], and it included label and

pectoral muscle removal, use of median filter, CLAHE, unsharp masking and extracting suspicious regions using K-means clustering.

Haze-Reduced Adaptive Technique (HRAT) is proposed as an improved contrast enhancement technique by Chakravarthy et al. [12]. The algorithm for the HRAT approach consists of haze removal based on the dark channel prior method, adaptive global-local transformations, integration of haze removal and adaptive transformation and finally repetition of the above steps for all pixels and channels.

Filtering is used as a technique to smooth, upgrade and distinguish the edges in an image. Median filter, mean filter, adaptive mean filter, Wiener filter and Gaussian filter are the most used filtering methods in mammogram image enhancement [13]. A method based on filtering techniques is used to improve the quality of mammogram images in [14]. It used the spatial averaging filter followed by histogram-based thresholding for image enhancement and noise reduction. A study by Mahmood et al. [15] used preprocessing steps such as adaptive median filtering, Gaussian filtering and bilateral filtering along with CLAHE and wavelet analysis for enhancing micro-calcifications in mammograms. Difference of Gaussian filter and Hessian of Gaussian method are used as de-noising and enhancement approaches for micro-calcifications in mammograms by Ali et al. [16]. The Laplacian filter is used as an image enhancement method in the study by Hossain et al. [17] for detecting calcifications.

In order to address the limitations of the existing enhancement methods for mammograms, this study proposes a dual-path enhancement pipeline specifically designed for the enhancement of breast masses and calcifications in mammogram images.

III. METHODOLOGY

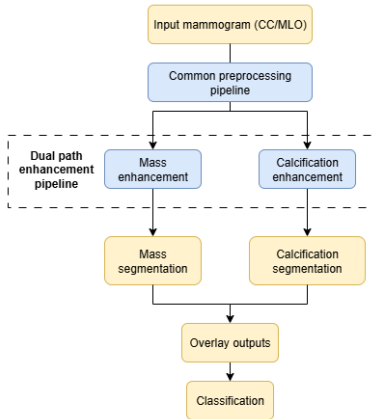


Fig. 1: The complete workflow of the CAD system

Fig. 1 shows the complete workflow of the CAD system and the dual path enhancement pipeline. The proposed image enhancement pipeline focuses on enhancing masses and calcifications of mammograms to obtain an improved visualisation. The pipeline was evaluated using the publicly available INBreast dataset [18], which provides 412 high-resolution mammograms in the Digital Imaging and

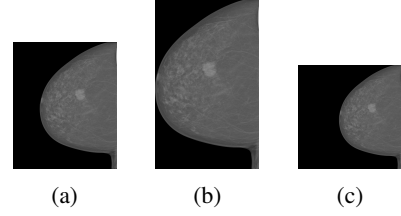


Fig. 2: Common preprocessing steps for masses and calcifications (a) Original image (b) Cropped and resized image (c) Standardised image

Communications in Medicine (DICOM) format along with annotations for Regions of Interest (ROI).

A. Preprocessing

The mammogram images from the dataset, originally in the DICOM format, were converted to Portable Network Graphics (PNG) format before processing. The dataset contained mammograms from both CC and MLO views.

The pectoral muscle in the MLO view images can cause misdiagnosis and affect the segmentation and classification [19]. Therefore, the MLO view images were filtered out, and the pectoral muscle was removed using the pectoral muscle annotations.

The INBreast dataset consists of images with two resolutions, namely 3328 x 4084 pixels and 2560 x 3328 pixels. In order to standardise the size of these images few steps were followed.

Initially, Otsu's adaptive thresholding [20] method was applied to separate foreground tissue from the background. Then, after binarisation, the external contours of the breast tissue region were identified using the contour detection algorithm in OpenCV. The largest contour, which corresponds to the primary breast tissue region, was identified, and a bounding rectangle was created around this contour. The cropped mammogram was obtained by cropping the original mammogram according to the dimensions of the bounding rectangle obtained above.

Image cropping was followed by a proportional resizing approach, keeping the aspect ratio of the image as a constant. The target size of the resizing was 1024 x 1024 pixels and was selected based on the requirement to preserve the important details of mammograms and the computational power required in further analysis. The resizing algorithm first determined if the image was portrait or landscape in orientation. Based on this, a scaling factor was calculated using the target size as the reference dimension. The scaling factor was used such that the primary dimension was standardised to the target size, while the secondary dimension was scaled proportionally to avoid distortions.

Following the proportional resizing, a zero-intensity padding based on the laterality was done to obtain the target size. These preprocessing steps resulted in a dataset of standardised mammograms preserving the original anatomical structures and the aspect ratios by avoiding any distortions. The images obtained through the preprocessing are shown in Fig. 2.

Masses and calcifications in mammograms have significant structural and visual differences. Therefore, a single enhancement method was insufficient to enhance both masses and calcifications. Accordingly, this approach proposes two image enhancement pipelines for mammograms, where each pipeline focuses on enhancing the visualisation of each abnormality. The final standardised mammogram images from the preprocessing were used as input for the enhancement of masses and calcifications.

B. Enhancement of Masses

In the mass enhancement pipeline, the standardised images first underwent contrast enhancement using CLAHE. Experimental values were used for the parameters clip limit and tile grid size.

Following contrast enhancement, a global threshold was used to highlight the suspicious regions. The threshold value was decided through experiments.

To further enhance the visualisation, magma colour mapping was used. Fig. 3 shows the output images after the initial steps of the mass enhancement pipeline. This converted the greyscale mammograms to a multi-channel representation where subtle intensity variations were mapped to different colour gradients. Afterwards, the colour-mapped images were decomposed into their constituent red (R), green (G), and blue (B) channels and fused images were obtained by the fusion of two selected colour channels. Finally, three fused images were obtained as red-green (R-G) fused, red-blue (R-B) fused and blue-green (B-G) fused images.

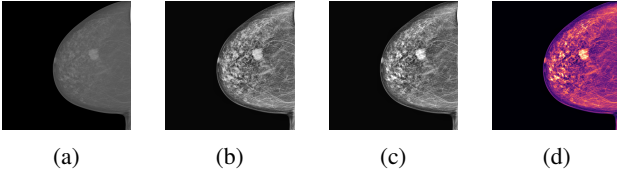


Fig. 3: Output images of initial steps of mass enhancement pipeline (a) Standardised image (b) CLAHE applied image (c) Threshold-based intensity enhanced image (d) Magma colour map visualisation of thresholded image

The multi-channel decomposition and selective fusion approach enabled enhanced visualisation of ROI, allowing enhancement of breast masses as shown in Fig. 4, and Fig. 5.

C. Enhancement of Calcifications

This pipeline is specifically designed to enhance the visualisation of calcifications in mammograms. First, a Gaussian low-pass filter was applied to the standardised greyscale mammogram. This resulted in a smoothed image with low-frequency tissue structures. The smoothed image was then subtracted from the standardised mammogram to extract the high-frequency components. These two steps collectively create a high-pass filtering technique. Afterwards, a rectified image was obtained by setting all negative intensity values to zero.

A secondary Gaussian smoothing was performed on the rectified image to improve the signal-to-noise ratio

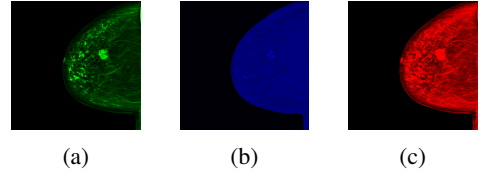


Fig. 4: Images obtained by channel separation (a) Green channel (b) Blue channel (c) Red channel

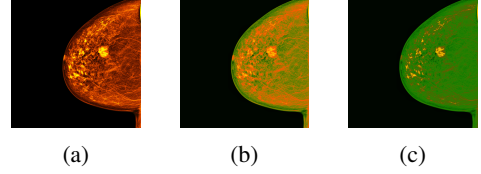


Fig. 5: Images obtained by channel fusion (a) Fused R-G image (b) Fused R-B image (c) Fused B-G image

(SNR) and was followed by an adaptive threshold step to further enhance the calcifications. Finally, a binary mask of suspicious calcifications was created.

In order to obtain an improved visualisation, the obtained binary mask was overlaid on the standardised mammogram. This combination provided an image which consists of the suspicious calcifications prominently highlighted against the anatomical background, and the results are shown in Fig. 6.

D. Evaluation metrics

The following parameters were used for evaluating the performance of approaches for the enhancement of both masses and calcifications.

- 1) Contrast-to-Noise ratio (CNR) : CNR measures the contrast between the ROI (mass or calcification) and the background [21].

$$CNR = \frac{\mu_{\text{signal}} - \mu_{\text{bg}}}{\sigma_{\text{bg}}} \quad (1)$$

- μ_{signal} = mean intensity of the ROI,
- μ_{bg} = mean intensity of the background,
- σ_{bg} = standard deviation (SD) of the background

- 2) Signal-to-Background ratio (SBR) : SBR measures the relative brightness of the ROI compared to the background.

$$SBR = \frac{\mu_{\text{signal}}}{\mu_{\text{bg}}} \quad (2)$$

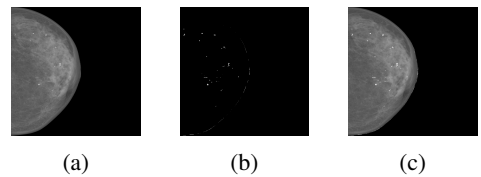


Fig. 6: Output images of enhancement pipeline for calcifications (a) Standardised image (b) Binary mask obtained by adaptive thresholding (c) Final enhanced image

TABLE I: Comparison of Enhancement Methods Across Evaluation Metrics for Masses

Method	CNR (mean $\pm$ SD)	SBR (mean $\pm$ SD)	Local Contrast (mean $\pm$ SD)	SSIM Local (mean $\pm$ SD)	SSIM Global (mean $\pm$ SD)	Entropy (mean $\pm$ SD)	Colorfulness (mean $\pm$ SD)
CLAHE	2.38 $\pm$ 0.57	4.30 $\pm$ 1.21	30.47 $\pm$ 13.62	0.46 $\pm$ 0.13	0.25 $\pm$ 0.03	3.18 $\pm$ 0.66	nan $\pm$ nan
HRAT	2.44 $\pm$ 0.75	4.30 $\pm$ 1.43	39.69 $\pm$ 14.22	0.30 $\pm$ 0.08	0.51 $\pm$ 0.06	2.40 $\pm$ 0.45	nan $\pm$ nan
Magma colour map	1.99 $\pm$ 0.53	4.12 $\pm$ 1.49	14.95 $\pm$ 12.42	0.39 $\pm$ 0.04	0.69 $\pm$ 0.01	2.97 $\pm$ 0.59	72.03 $\pm$ 7.75
R-G Fusion	2.74 $\pm$ 0.72	5.56 $\pm$ 1.92	42.83 $\pm$ 16.77	0.52 $\pm$ 0.09	0.53 $\pm$ 0.07	3.22 $\pm$ 0.66	70.30 $\pm$ 8.32
R-B Fusion	2.04 $\pm$ 0.52	3.76 $\pm$ 1.14	24.99 $\pm$ 10.87	0.37 $\pm$ 0.06	0.51 $\pm$ 0.06	2.65 $\pm$ 0.51	98.42 $\pm$ 11.40
B-G Fusion (Proposed)	4.10 $\pm$ 0.74	5.93 $\pm$ 2.11	46.58 $\pm$ 17.96	0.91 $\pm$ 0.06	0.97 $\pm$ 0.08	3.63 $\pm$ 0.72	102.08 $\pm$ 12.49

TABLE II: Comparison of Enhancement Methods Across Evaluation Metrics for Calcifications

Method	CNR (mean $\pm$ SD)	SBR (mean $\pm$ SD)	Local Contrast (mean $\pm$ SD)	PSNR (mean $\pm$ SD)	SSIM Global (mean $\pm$ SD)	SSIM Local (mean $\pm$ SD)
CLAHE	2.28 $\pm$ 0.62	4.06 $\pm$ 1.14	9.14 $\pm$ 27.69	29.90 $\pm$ 0.28	0.25 $\pm$ 0.03	0.48 $\pm$ 0.09
Gaussian filter	1.71 $\pm$ 0.48	2.44 $\pm$ 1.13	1.04 $\pm$ 4.47	32.92 $\pm$ 1.36	0.80 $\pm$ 0.02	0.71 $\pm$ 0.12
Laplacian filter	0.99 $\pm$ 0.57	2.40 $\pm$ 0.84	1.00 $\pm$ 0.06	28.07 $\pm$ 0.09	0.03 $\pm$ 0.01	-0.04 $\pm$ 0.02
Proposed method	2.30 $\pm$ 0.82	4.41 $\pm$ 1.60	1.06 $\pm$ 0.62	43.66 $\pm$ 1.60	0.96 $\pm$ 0.01	0.88 $\pm$ 0.09

- 3) Local contrast (σ) : Local contrast measures the contrast in the ROI.

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (3)$$

- N = number of pixels in the ROI,
- x_i = intensity value of the i^{th} pixel in the ROI,
- μ = mean intensity value of all pixels in the ROI.

- 4) Structural Similarity Index Global (SSIM Global and SSIM Local) : SSIM global measures structural similarity between whole enhanced image and the original image. In order to calculate SSIM for the enhanced image with masses, the average value of SSIM per R,G,B channels is obtained. Local SSIM measures structural similarity considering the ROI [22].

The following parameters were used as evaluation parameters specific for mass enhancement.

- 1) Entropy ($H(X)$) : Entropy measures the richness of the pixel intensity distribution [23].

$$H(X) = - \sum_{i=1}^N p(x_i) \log_2 p(x_i) \quad (4)$$

- $p(x_i)$ = probability of occurrence of intensity x_i ,
- N = total number of unique pixel intensity values.

- 2) Colourfulness : Colourfulness measures how rich the colours are and is calculated using Haster and Susstrunk formulae.

$$\text{Colorfulness} = \sqrt{\sigma_{rg}^2 + \sigma_{yb}^2} + 0.3 \times \sqrt{\mu_{rg}^2 + \mu_{yb}^2} \quad (5)$$

- σ_{rg} = SD of the R-G channel difference,

- σ_{yb} = SD of the yellow-blue(Y-B) channel difference,
- μ_{rg} = the mean of the R-G channel difference,
- μ_{yb} = the mean of the Y-B channel difference.

The following parameters were used as evaluation parameters specific for calcification enhancement.

- 1) Peak Signal-to-Noise ratio (PSNR) : PSNR Compares enhanced image with original image to assess distortion [24].

$$\text{PSNR} = 10 \times \log_{10} \left(\frac{\text{MAX}_{\text{pixel}}^2}{\text{MSE}} \right) \quad (6)$$

- $\text{MAX}_{\text{pixel}}$ = the maximum possible pixel value
- MSE = the mean squared error between the original and the processed image.

IV. RESULTS AND DISCUSSION

This section presents the results of the evaluation of the proposed enhancement pipelines using the evaluation metrics stated above.

A. Results for Enhancement of Masses

The proposed enhancement pipeline for breast masses was compared with a few existing methods for mammogram enhancement, namely, CLAHE, HRAT, and Magma colour mapping and other fusion strategies such as R-G and R-B fusion. The results are summarised in Table I. The nan values indicate that the considered metric is not applicable for the enhanced image due to the image properties. According to the results, the proposed B-G fused image obtained well-balanced results across all metrics.

B. Results for Enhancement of Calcifications

The proposed pipeline for enhancement of calcifications was evaluated by comparing its performance with the existing methods such as CLAHE, Gaussian and Laplacian filtering. The results are summarised in Table II.

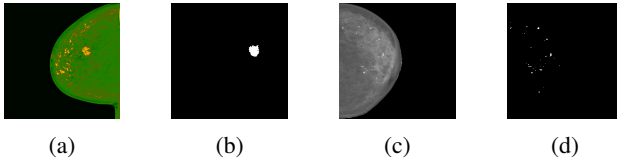


Fig. 7: Comparison of enhanced masses and calcifications with original annotations (a) Final enhanced image (with mass) (b) Original annotation (mass) (c) Final enhanced image (with calcifications) (d) Original annotation (calcifications)

The proposed enhancement pipeline for breast calcifications achieved the highest values for PSNR, SSIM local and global, and this indicates better noise suppression and preservation of structural information. Fig. 7 shows the comparison of enhanced masses and calcifications with the original annotation of the ROI.

V. CONCLUSION

The study proposed a dual-path enhancement pipeline specifically designed for the enhancement of breast masses and calcifications in mammogram images. According to the quantitative evaluation, the proposed methods for enhancement of masses and calcifications yielded better results when compared with the conventional enhancement approaches used for mammogram images. The improvements represented by the results of the evaluation metrics demonstrate that this enhancement pipeline can facilitate the detection of breast cancer more accurately. As the proposed pipeline is evaluated only using a single dataset, the generalizability of the proposed approach should be tested using other datasets as well. Future work explores the potential of using these enhanced images in DL based CAD systems to further automate and optimise the detection of breast cancer with better performance in breast mass and calcification segmentation and classification into BI-RADS categories [25] [26]. In conclusion, this dual-path enhancement pipeline has the potential to assist in overcoming challenges of visualising breast masses and calcifications in mammograms.

REFERENCES

- [1] World Health Organization, "Breast cancer," 2024, accessed: Dec. 19, 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/breast-cancer>
- [2] National Breast Cancer Foundation, "Early detection," 2024, accessed: Mar. 25, 2025. [Online]. Available: <https://www.nationalbreastcancer.org/early-detection-of-breast-cancer/>
- [3] American Cancer Society, "Breast cancer early detection and diagnosis," <https://www.cancer.org/cancer/types/breast-cancer/screening-tests-and-early-detection.html>, 2024, [Online; Accessed: Mar. 25, 2025].
- [4] C. Lamprou, K. Katsikari, N. Rahmani, L. J. Hadjileontiadis, M. Seghier, and A. Alshehhi, "Stethonet: Robust breast cancer mammography classification framework," *IEEE Access*, 2024.
- [5] D. Shen, G. Wu, and H.-I. Suk, "Deep learning in medical image analysis," *Annual review of biomedical engineering*, vol. 19, no. 1, pp. 221–248, 2017.
- [6] Y. E. Almalki, T. A. Soomro, M. Irfan, S. K. Alduraibi, and A. Ali, "Impact of image enhancement module for analysis of mammogram images for diagnostics of breast cancer," *Sensors*, vol. 22, no. 5, p. 1868, 2022.
- [7] M. Biltawi, N. Al-Najdawi, and S. Tedmori, "Mammogram enhancement and segmentation methods: classification, analysis, and evaluation," in *The 13th international Arab conference on information technology*, 2012.
- [8] Y. V. Nalawade, "Evaluation of breast calcifications," *Indian Journal of radiology and imaging*, vol. 19, no. 04, pp. 282–286, 2009.
- [9] E. D. Pisano, S. Zong, B. M. Hemminger, M. DeLuca, R. E. Johnston, K. Muller, M. P. Braeuning, and S. M. Pizer, "Contrast limited adaptive histogram equalization image processing to improve the detection of simulated spiculations in dense mammograms," *Journal of Digital imaging*, vol. 11, pp. 193–200, 1998.
- [10] M. Zeng, Y. Li, Q. Meng, T. Yang, and J. Liu, "Improving histogram-based image contrast enhancement using gray-level information histogram with application to x-ray images," *Optik*, vol. 123, no. 6, pp. 511–520, 2012.
- [11] H. Avci and J. Karakaya, "A novel medical image enhancement algorithm for breast cancer detection on mammography images using machine learning," *Diagnostics*, vol. 13, no. 3, p. 348, 2023.
- [12] S. Chakravarthy, B. Nagarajan, V. V. Kumar, T. Mahesh, R. Sivakami, and J. R. Annand, "Breast tumor classification with enhanced transfer learning features and selection using chaotic map-based optimization," *International Journal of Computational Intelligence Systems*, vol. 17, no. 1, p. 18, 2024.
- [13] S. Mudrakola and N. Hegde, "Removal of noise on mammogram breast images using filtering methods," *Concurrency and Computation: Practice and Experience*, vol. 35, no. 1, p. e7444, 2023.
- [14] A. J. Méndez, P. G. Tahoces, M. J. Lado, M. Souto, J. Correa, and J. J. Vidal, "Automatic detection of breast border and nipple in digital mammograms," *Computer methods and programs in biomedicine*, vol. 49, no. 3, pp. 253–262, 1996.
- [15] T. Mahmood, J. Li, Y. Pei, F. Akhtar, A. Imran, and M. Yaqub, "An automatic detection and localization of mammographic microcalcifications roi with multi-scale features using the radiomics analysis approach," *Cancers*, vol. 13, no. 23, p. 5916, 2021.
- [16] M. Alsheh Ali, M. Eriksson, K. Czene, P. Hall, and K. Humphreys, "Detection of potential microcalcification clusters using multivendor for-presentation digital mammograms for short-term breast cancer risk estimation," *Medical physics*, vol. 46, no. 4, pp. 1938–1946, 2019.
- [17] M. S. Hossain, "Microcalcification segmentation using modified u-net segmentation network from mammogram images," *Journal of King Saud University-Computer and Information Sciences*, vol. 34, no. 2, pp. 86–94, 2022.
- [18] S. P. Ramanathan, "Inbreast dataset," <https://www.kaggle.com/datasets/ramanathanp20/inbreast-dataset>, 2020, accessed: 2025-04-27.
- [19] S. Rahimeto, T. G. Debele, D. Yohannes, and F. Schwenker, "Automatic pectoral muscle removal in mammograms," *Evolving Systems*, vol. 12, pp. 519–526, 2021.
- [20] D. Liu and J. Yu, "Otsu method and k-means," in *2009 Ninth International conference on hybrid intelligent systems*, vol. 1. IEEE, 2009, pp. 344–349.
- [21] A. H. Baydush and C. E. Floyd Jr, "Improved image quality in digital mammography with image processing," *Medical physics*, vol. 27, no. 7, pp. 1503–1508, 2000.
- [22] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE transactions on image processing*, vol. 13, no. 4, pp. 600–612, 2004.
- [23] C. E. Shannon, "A mathematical theory of communication," *ACM SIGMOBILE mobile computing and communications review*, vol. 5, no. 1, pp. 3–55, 2001.
- [24] P. Kellman and E. R. McVeigh, "Image reconstruction in snr units: a general method for snr measurement," *Magnetic resonance in medicine*, vol. 54, no. 6, pp. 1439–1447, 2005.
- [25] T. D. Wickramasingha, S. A. A. J. M. Athukorala, S. M. D. Deepashika, I. A. S. I. Fernando, U. L. Wijewardhana, and U. B. Balagalla, "A unified deep learning approach for the segmentation of breast masses and calcifications," in *Proceedings of the Moratuwa Engineering Research Conference (MERCon)*, 2025, in press.
- [26] S. M. D. Deepashika, T. D. Wickramasingha, S. A. A. J. M. Athukorala, M. S. Induwara, U. L. Wijewardhana, and U. B. Balagalla, "Multi view mammogram analysis with segmented breast masses and calcifications for bi-rads classification," unpublished.